

Keycloak Authentication — ONE-STOP LAB GUIDE

Keep this guide handy in lab. Covers all 3 situations.

DOWNLOAD LINKS (do at home before lab)

What	Where to Download	Used In
Keycloak ZIP	keycloak.org/downloads → ZIP	Windows (home practice)
Keycloak TAR.GZ	keycloak.org/downloads → TAR.GZ	Ubuntu when DNS fails
Java JDK 17+	adoptium.net	If Java not installed
jwt.io	browser — no download	Decode token

WHICH METHOD TO USE?

Situation in Lab	Use Method
Ubuntu + Docker working + internet OK	Method A: Docker
Ubuntu + DNS error / quay.io fails	Method B: TAR file on Ubuntu
Windows laptop at home	Method C: ZIP on Windows

METHOD A: Ubuntu + Docker (Original)

Use when internet/ethernet is working fine in college.

1. Install Docker:

```
sudo apt install docker.io -y
sudo systemctl start docker
```

2. Pull Keycloak:

```
sudo docker pull quay.io/keycloak/keycloak:latest
```

3. Run container:

```
sudo docker run -d -p 8080:8080 -e KEYCLOAK_ADMIN=admin -e
KEYCLOAK_ADMIN_PASSWORD=admin123 quay.io/keycloak/keycloak:latest start-dev
```

4. Check running:

```
sudo docker ps
```

Open browser: <http://localhost:8080> → Login: admin / admin123

METHOD B: Ubuntu + TAR File (DNS Error Fallback)

Use when DNS error or docker pull quay.io fails. Bring TAR.GZ on pen drive!

1. Copy keycloak-26.6.3.tar.gz to Ubuntu desktop

2. Open terminal in that folder and extract:

```
tar -xvzf keycloak-26.6.3.tar.gz
```

3. Go inside and start:

```
cd keycloak-26.6.3/bin  
./kc.sh start-dev
```

Wait 1-2 min for: 'Keycloak started' in terminal

Open browser: <http://localhost:8080> → Create admin: admin / admin123

METHOD C: Windows Laptop (Home Practice)

No Docker or WSL needed! Just Java + Keycloak ZIP.

1. Check Java: java -version (need 17+)

2. Extract ZIP to C:\keycloak

Right click ZIP → Extract All → choose C:\keycloak

3. Open CMD in C:\keycloak\bin folder:

Click address bar → type cmd → Enter

```
kc.bat start-dev
```

Open browser: <http://localhost:8080> → Create admin: admin / admin123

KEYCLOAK CONFIGURATION (Same for ALL Methods)

Step 1: Create Realm

Top-left dropdown → Create Realm → Name: OnlineExamRealm → Create

Step 2: Create Client

Left menu → Clients → Create Client

Client ID: exam-app | Client Type: OpenID Connect → Next

Turn ON: Direct Access Grants → Save (leave Root URL and Home URL blank)

Step 3: Create Roles

Left menu → Realm Roles → Create Role → Name: student → Save

Repeat for: faculty and admin

Step 4: Create User

Left menu → Users → Add User

Username: student1 | First name: Student | Last name: One

Email: student1@test.com | Email Verified: ON → Create

Step 5: Set Password

Click Credentials tab → Set Password → Password: 1234 → Temporary: OFF → Save

Step 6: Assign Role

Click Role Mapping tab → Assign Role → search student → check → Assign

GENERATE JWT TOKEN

Run this in terminal (Ubuntu) or CMD (Windows):

```
curl -X POST
http://localhost:8080/realms/OnlineExamRealm/protocol/openid-connect/token -H
"Content-Type: application/x-www-form-urlencoded" -d
"client_id=exam-app&username=student1&password=1234&grant_type=password"
```

Copy the `access_token` value (long string starting with eyJ...)

Open <https://jwt.io> → paste token in LEFT box → right side shows username + role + expiry ■

JAVA FILES SETUP AND RUN

Create folder IS_exp8 → put ExamClientGUI.java, ExamServerGUI.java, DESUtil.java inside

Open CMD in that folder and compile:

```
javac *.java
```

Terminal 1 — Run Server:

```
java ExamServerGUI
```

→ Server GUI window opens, shows: Exam Server Started...

Terminal 2 — Run Client:

```
java ExamClientGUI
```

→ Login GUI opens. Enter: student1 / 1234 / any semester / any course code → Login & Start Exam

→ Question paper appears in the GUI window ■

COMMON ERRORS AND FIXES

Error	Fix
DNS error / quay.io fails	Use Method B — TAR file on pen drive
WSL not installed (Windows)	Use Method C — no Docker needed on Windows
Address already in use: 5000	Press Ctrl+C in old server terminal, then rerun
Account is not fully set up	Users → student1 → add name + email → Email Verified ON → Save
invalid_grant in curl	Check spelling: OnlineExamRealm (capital O, E, R)
Authentication Failed in GUI	Make sure Keycloak is running and password is 1234

QUICK CHECKLIST BEFORE SHOWING TO SIR

- [] Keycloak running at <http://localhost:8080>
- [] Realm: OnlineExamRealm created
- [] Client: exam-app created (Direct Access Grants ON)
- [] Roles created: student, faculty, admin
- [] User: student1 / 1234 created (Email Verified ON, role student assigned)
- [] curl command gives access_token
- [] Token decoded at jwt.io shows username + student role
- [] ExamServerGUI running on port 5000
- [] ExamClientGUI login works and shows question paper

Result: Secure user authentication successfully implemented using Keycloak with JWT token validation and role-based access control.